

Lecture 5: Risk Assessment Workshop

Disclaimer

The “project” presented in this work is for educational purposes only and does not refer to any actual person or organization.

In other words, pretend this happened on Planet Astra.

Executive Summary

There are 20,000 secondary school students in a province (77 schools).

The education committee of the province wants us (or someone) to help them survey and collect the mental health information (PHQ-A form) from these students.

The committee has two conflicting goals:

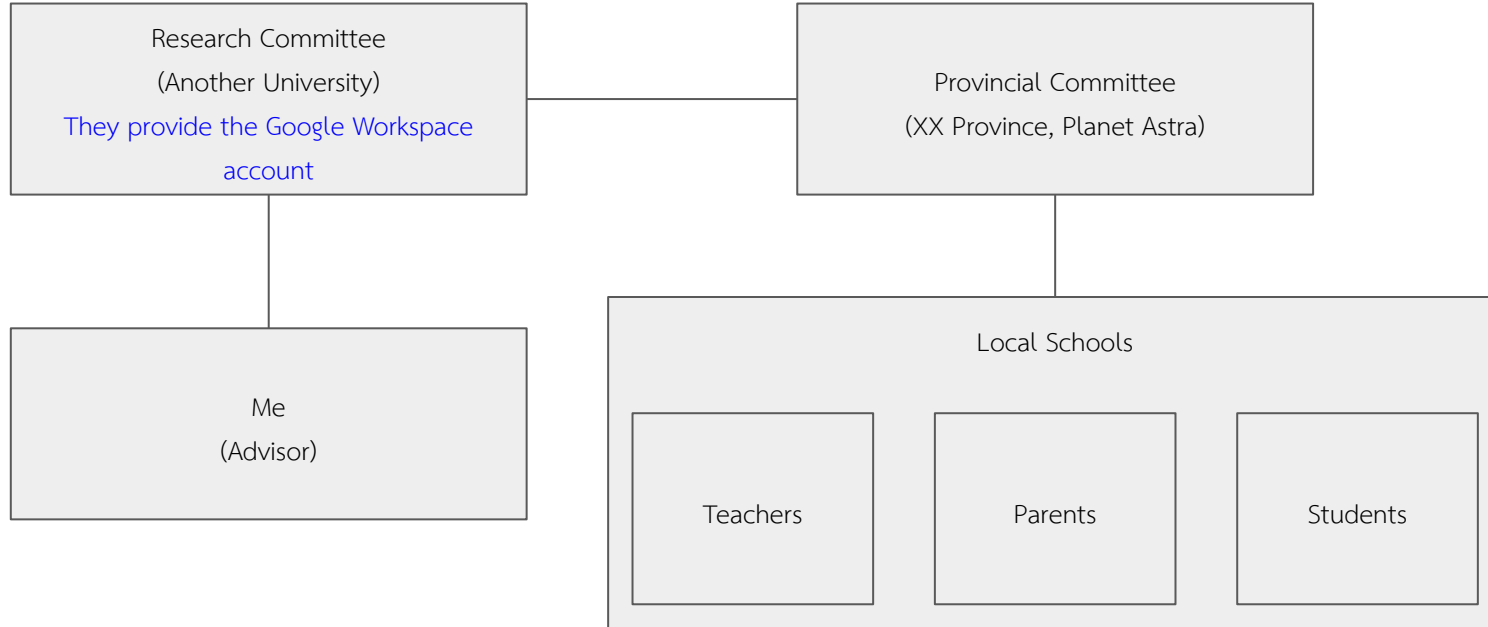
- One is to perform a survey “out of 20,000 students, 5,000 are at risk of ...”
 - No need to track individuals
- Another is to “help individual children”
 - Now we have to track individual students

Naive Database Looking for Trouble

PIIs

UID	First Name	Last Name	School	Grade	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	
					1	2	0	2	2	3	2	1	0	

People involved



Consent Form

What do you need to know before you can write a consent form?

- Age of Data Subject
- Local Law (Jurisdiction)
- What data is being collected?
 - Must answer the goal
 - Also must be *minimal*
- The purpose of data collection?
- Retention Period (How long)

Consent of Children

Can parents consent on behalf of children?

Legally, yes.

The children cannot consent on their own.

Suppose we have an experiment.

Can parents “send” children to experiment? Legally, yes.

Therefore, the committee wants me to provide a consent form with ONLY parental consent. No child consent is required if the parent consented.

Excuses to get the children to participate in human subject research?

- Extra credit at school (there are benefits for the children)
- Benefits for the parents
- For a better world
- Experience for the children

1. **“It’s for science.”** — 28
2. **“It’ll help other children someday.”** — 22
3. **“They’ll get paid / get a gift card.”** — 16
4. **“It’s basically educational.”** — 12
5. **“The researchers say it’s perfectly safe.”** — 9
6. **“The parents consented, so it’s fine.”** — 7
7. **“They need kids for the study.”** — 4
8. **“It’ll look good / be a cool experience.”** — 2

Things to maintain in your mind when arguing:

- The rights of the children
- If you force children to participate anyway
 - Violation of child rights
 - Your data now has validity concerns => students can just mark the answers randomly, not sincerely cooperate with the tests, etc.
 -

Try asking AI this:

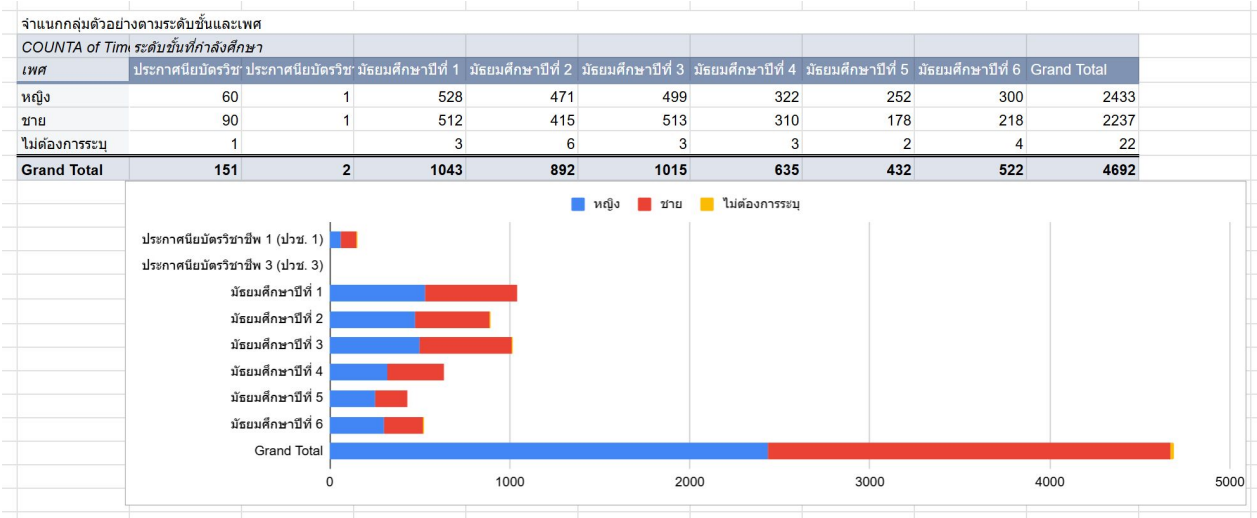
I want to ask about consent of children in some experiment. What if the "truly random" sampling of children is important, and consent will skew the sampling?

For example, I want to perform an experiment about the skills of local children at solving escape rooms. I don't want only consenting children to participate, because then it means the cohort will be self-selected and could be biased. Therefore, I have an ethical excuse to use random sampling and kidnap the children to my escape room.

Argue for or against.

Pseudonymization

UUID	Gender	Grade	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	
	F	m5	1	2	0	2	2	3	2	1	0	
	X	v3	3	2	1	2	1	1	1	0	0	



Hunt the Witch (thought experiment)

A simple thought experiment.

Assume that there is a specific target you want to find in the dataset.

How do you find the Witch?

Example of Reidentification

Name	ID	Gender	Zip Code
T	4831	M	10900
O	8821	M	10900
H	9911	M	10800

Voter Registration
(Leaked from ECA Electoral
Commission of Astra)

Insurance Dataset (Public)

Gender	Zip Code	Disease
F	10400	Skin Cancer
M	10800	Testicular Cancer

Jurisdiction Risk

- Marketing targeting the EU (GDPR)
- Data crossing borders

Unclear Role

Role	Meaning	Simple Example	Context of this Project
Data Controller	Decides why personal data is collected and how it will be used.	A company decides to collect customer data for marketing.	Research Committee
DPO (Data Protection Officer)	Oversees an organization's data protection and privacy compliance , advises on obligations, and often serves as a contact point for individuals and regulators.	A company's DPO reviews privacy practices, advises teams on data-protection requirements, and handles privacy-related inquiries.	The Research Committee University's DPO
Data Collector	Collects or obtains data from the data subject or another source. This is often a functional description rather than a formally defined legal role.	A website collects names and email addresses through a signup form.	Teacher
Data Processor	Processes personal data on behalf of and under the instructions of the controller .	A cloud provider stores customer data for a company.	
Data Recipient	A person or organization to whom personal data is disclosed or provided .	An insurer receives employee information from an employer.	
Data Subject	The individual whom the personal data is	A customer whose name	